

DATA DICTIONARY

Customer Engagement & Product Utilization Analytics

European Banking Customer Dataset • 10,000 Records • 14 Variables

Prepared by
Manyu

B.Tech Information Technology
Unified Mentor – Data Analytics Internship

Dataset Reference & Analytical Metadata

1. Dataset Overview

This Data Dictionary documents the variables used in the **Customer Engagement & Product Utilization Analytics for Retention Strategy** project. It defines the meaning, type, analytical role, and interpretation notes for each source field.

The cleaned dataset contains **10,000 customer records and 14 columns**. Validation identified **0 missing values, 0 duplicate rows, and 0 duplicate Customer IDs**. The supplied Year field contains 2025 only.

Purpose: support analysis of customer engagement, product utilization, financial commitment, and churn-related behavior for retention strategy.

2. Dataset Summary

Property	Value
Records	10,000
Columns	14
Year	2025 only
Missing values	0
Duplicate rows	0
Duplicate Customer IDs	0
CustomerId	Unique identifier

3. Complete Data Dictionary

Definition and analytical role of every source variable:

Field	Type	Meaning	Analytical Role	Note
Year	Integer	Dataset year	Time/context	2025 only in supplied data.
CustomerId	Integer / ID	Unique customer identifier	Customer key	Unique customer reference.
Surname	Text	Customer surname	Descriptive	Not used as a retention KPI.
CreditScore	Integer	Customer credit score	Profile	Credit-profile context.
Geography	Categorical	Customer country/region	Segmentation	Supports geographic filtering.
Gender	Categorical	Customer gender	Segmentation	Supports segment filtering.
Age	Integer	Customer age	Profile	Demographic context.
Tenure	Integer	Years with bank	Relationship duration	Relationship context.
Balance	Numeric	Account balance	Financial commitment	Used for high-balance analysis.
NumOfProducts	Integer	Number of products used	Product utilization	Core product-depth variable.
HasCrCard	Binary 0/1	Credit card ownership	Relationship indicator	Used for stickiness analysis.
IsActiveMember	Binary 0/1	Activity indicator	Engagement	Core engagement variable.
EstimatedSalary	Numeric	Estimated salary	Financial profile	Used for financial segmentation.
Exited	Binary 0/1	Churn indicator	Outcome	1 = churned; 0 = retained.

4. Variable Classification

Category	Variables	Purpose
Identifier	CustomerId	Unique customer reference.
Categorical	Surname, Geography, Gender	Description and segmentation.
Continuous numeric	Balance, EstimatedSalary	Financial context.
Discrete numeric	Year, CreditScore, Age, Tenure, NumOfProducts	Profile, tenure and product-depth analysis.
Binary	HasCrCard, IsActiveMember, Exited	Product ownership, engagement and churn.

5. Derived Dashboard Fields

Derived Field	Definition	Purpose
Activity Status	Active if IsActiveMember = 1; otherwise Inactive.	Readable engagement status.
Churn Status	Retained if Exited = 0; Churned if Exited = 1.	Readable customer status.
Risk Score	Weighted score using inactivity, low product depth, high balance and churn signals.	Exploratory risk segmentation.
Risk Level	Score mapped to Low, Medium or High Risk.	Dashboard risk monitoring.

Important: the project-defined Risk Score includes the Exited outcome itself, so it should be treated as an exploratory segmentation score, not a predictive churn model.

6. Required KPI Variables

KPI	Main Variables	Meaning
Engagement Retention Ratio	IsActiveMember + Exited	Compares active and inactive retention.
Product Depth Index	NumOfProducts + Exited	Relates product depth to retention.
High-Balance Disengagement Rate	Balance + IsActiveMember	Inactivity among customers at/above €100,000.
Credit Card Stickiness Score	HasCrCard + Exited	Retention difference by card ownership.
Relationship Strength Index	IsActiveMember + NumOfProducts	Combined engagement/product score on project scale.

7. Data Quality & Validation

- **Completeness:** 0 missing values.
- **Uniqueness:** 0 duplicate rows and 0 duplicate Customer IDs.
- **Year coverage:** 2025 only; the dataset is not a multi-year trend dataset.
- **Product count:** NumOfProducts ranges from 1 to 4.
- **Balance threshold:** €100,000 is a project exploration threshold, not a universal banking standard.

8. Dashboard Filters

Filter	Field	Purpose
Geography	Geography	Regional exploration.
Gender	Gender	Gender-based segmentation.
Number of Products	NumOfProducts	Product-depth filtering.
Balance Range	Balance	Financial-value segmentation.
Salary Range	EstimatedSalary	Salary-based exploration.
Engagement Status	Activity Status	Active/inactive filtering.
Customer Status	Churn Status	Retained/churned filtering.

9. Interpretation Guidelines

- IsActiveMember is the dataset's activity indicator; it should not automatically be treated as a complete measure of engagement across every channel.
- NumOfProducts measures product count, not product profitability or usage intensity.
- Balance represents account balance and should not automatically be treated as total customer value.
- EstimatedSalary is an estimated financial attribute and should be treated as contextual.
- Exited records churn status but does not provide the reason for churn.
- Observed relationships are descriptive associations within this dataset and should not be treated as proof of causation.

10. Dataset-to-Dashboard Mapping

Dashboard Area	Inputs	Purpose
Customer Overview	Exited, IsActiveMember, Balance, NumOfProducts	Overall customer and retention indicators.
Engagement Analysis	IsActiveMember, Exited	Engagement vs churn analysis.
Product Utilization	NumOfProducts, Exited, IsActiveMember, Balance	Product-depth analysis.
High-Balance Disengagement	Balance, IsActiveMember	High-value inactive segment monitoring.
Credit Card Analysis	HasCrCard, Exited	Retention by card ownership.
Risk Explorer	IsActiveMember, NumOfProducts, Balance, Exited	Exploratory risk segmentation.

11. Technical Notes & Project Information

The source file used by the Streamlit dashboard is **European_Bank_Cleaned.csv**. The application uses Pandas for data handling and Streamlit for interactive dashboard delivery.

Project: Customer Engagement & Product Utilization Analytics for Retention Strategy

Author: Manya

Program: B.Tech Information Technology

Internship: Unified Mentor – Data Analytics Internship

End of Data Dictionary